Compactness for natural deduction

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:proves-compact of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- natural deduction derivations and their undischarged assumptions
- finiteness of derivations as finite trees
- consistency defined by absence of a derivation of a contradiction

Proposition 0.1 (Compactness)**.** *1. If* $\Gamma \vdash !A$ *then there is a finite subset* $\Gamma_0 \subseteq \Gamma$ *such that* $\Gamma_0 \vdash !A$*.*

2. If every finite subset of Γ *is consistent, then* Γ *is consistent.*

Proof. 1. If $\Gamma \vdash !A$, then there is !!aderivation δ of $!A$ from Γ. Let Γ_0 be the set of !!undischarged assumptions of δ. Since any !!derivation is finite, Γ_0 can only contain finitely many !!sentences. So, δ is !!aderivation of $!A$ from a finite $\Gamma_0 \subseteq \Gamma$.

2. This is the contrapositive of (1) for the special case $!A\,\mathrm{id}\,\bot$.

□

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